NBA Video Generator

Ronen Huang

Objectives

1. Given any date, default is current, determine which players to make videos for based off their statlines - minutes, points (fg-fga, 3fg-3fga, ft-fta), rebounds (off, def), assists (to), steals, blocks, fouls, +\-.

2. Make video of players with the concatenation of play by play clips available from crawling the NBA website. Make a description text linking video time to play. Make relevant thumbnails with the most impressive statistics.

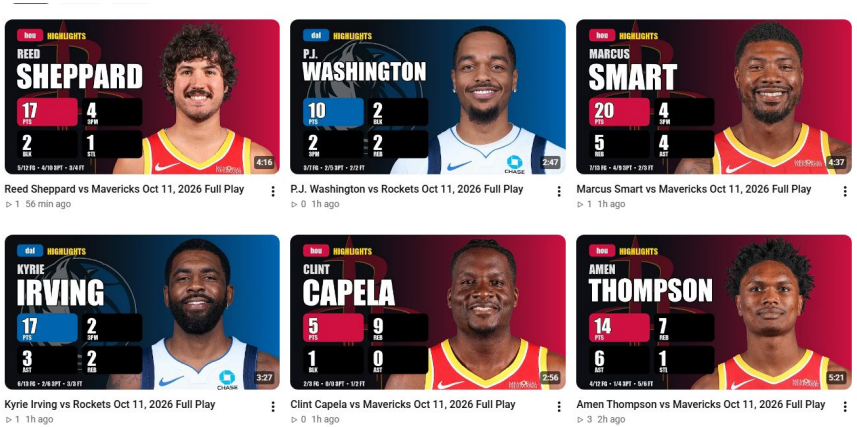
3. Upload the videos with relevant description and thumbnail to YouTube.

Product

YouTube: https://www.youtube.com/@NBAFullPlayHighlights/videos

PyPi: https://pypi.org/project/nba-video-generator/

Code: https://github.com/ronenh24/nba_video_generator



Manual Creation - pipeline

Given a list of tuples with player last name (according to PBP), date, and team abbreviation, crawls the matching play by play page on the NBA website to get play clips, merges plays within 5 seconds of each other, and concatenates to video. Makes description text linking video times to plays.

```
from nba_video_generator.beta_search import pipeline

pipeline(
    [
        ("DiVincenzo", "2026-04-20", "min"),
    ],
    {
        "ffmpeg_path": [ffmpeg path]
    }
)
```

Problems with Manual Creation

- If more than one player on a team shares the same last name (Franz Wagner, Moe Wagner), then last name must be abbreviated first name with last name which forces the user to look at the team roster.

- Since there are 30 teams in the league, there can be as much as 15 games a day which forces the user to look at many statlines.

- The algorithm never looks at the statlines and does not know the last name so thumbnails cannot be created.

Automatic Creation - nba_agent

Given a date, crawls the statlines of each game from the NBA website, determines the best statlines of each game with Qwen 3.5 LLM, makes videos and descriptions from the pipeline method with the list of last name, date, team tuples, makes the thumbnails for each video.

```
from nba_video_generator.run_agent import nba_agent

date = "2025-12-25"

ffmpeg_path = [ffmpeg path]

nba_agent(date, ffmpeg_path)
```

Further Improvements

- Remove duplicated clips
  - A play then the next play might share common video so the second instance should be deleted to ensure continuity.

- Compressed videos
  - Currently videos take up a lot of space to ensure quality.

Previous Version - Version 1

Crawled the statlines from the NBA website and obtained clips for specified events. Sorted by match to play by play. Unreliable compared to play by play.

```
from nba_video_generator.search import generate_video

jalen_green_assists_urls = generate_video(
    player_name="Jalen Green",
    date_start="2024-11-01",
    date_end="2024-11-08",
    team="hou",
    FGM=False,
    FGA=False,
    ThreePM=False,
    ThreePA=False,
    OREB=False,
    DREB=False,
    REB=False,
    AST=True,
    STL=False,
    BLK=False,
    TO=True,
    PF=False
)
```

```
from nba_video_generator.search import make_video

make_video(
    video_urls=jalen_green_assists_urls,
    base_name="jalen_green_assists",
    fps=30, preset="ultrafast",
    segment="Whole"
)
```

Previous Version - Version 2

Introduced a pipeline method to version 1 which takes a player parameters and video parameters for a list of players. Explored getting video from play by play.

```python
from nba_video_generator.search import pipeline

player_params = {
    "date_start": "2025-10-25",
    "date_end": "2025-10-25",
    "FGM": False,
    "FGA": True,
    "ThreePM": False,
    "ThreePA": False,
    "OREB": False,
    "DREB": False,
    "REB": True,
    "AST": True,
    "STL": True,
    "BLK": True,
    "TO": True,
    "PF": True,
    "include_ft": True
}

video_params = {
    "fps": 30,
    "preset": 'ultrafast',
    "segment": 'Play',
    "include_caption": True
}

name_team_base = [
    ("Kon Knueppel", "cha", "kon"),
    ("Joel Embiid", "phi", "embiid"),
    ("Bennedict Mathurin", "ind", "benn"),
    ("Cedric Coward", "mem", "cedric"),
    ("Javon Small", "mem", "javon"),
    ("Christian Braun", "den", "braun"),
    ("Devin Booker", "phx", "book")
]

pipeline(player_params, video_params, name_team_base)
```

```python
from nba_video_generator.beta_search import pipeline

pipeline(
    [
        ("Booker", "2026-02-19", "phx"),
    ]
)
```

Previous Version - Version 3

Made videos with the play by play from the NBA website rather than the unreliable statlines page. Downloaded plays with GUI (the window must be pressed).

```
from nba_video_generator.beta_search import pipeline

pipeline(
    [
        ("DiVincenzo", "2026-04-20", "min"),
    ],
    {
        "ffmpeg_path": r"C:\Users\ronen\Documents\Projects\nba_video_generator\src\nba_video
    }
)
```

```
from nba_video_generator.beta_search import pipeline

pipeline(
    [
        ("Booker", "2026-04-19", "2026-04-27", "phx"),
    ],
    {"ffmpeg_path": r"C:\Users\ronen\Documents\Projects\nba_video_generator\src\nba_video_
)
```

Previous Version - Version 4

Introduced automatic creation of videos with determination of best statlines.
Downloaded plays with GUI (the window must be pressed).

```
from nba_video_generator.run_agent import nba_agent

date = "2025-12-25"

ffmpeg_path = r"C:\Users\ronen\Documents\Projects\nb

nba_agent(date, ffmpeg_path)
```